

$$v = r\omega$$

$$8.4 = (16)\omega$$

$$\omega = 0.525 \text{ rad s}^{-1}$$

$$\theta = \omega t = (0.525)(8.0) = 4.2 \text{ rad}$$

Q8 Ans: D

Angular speed and the linear speed are different ways of describing the same motion; they do not cause each other.

By Newton's second law, net force causes acceleration and hence changes in both angular speed and the linear speed simultaneously.

Q8 Ans: C